

An Analysis of AlphaGo

AlphaGo

AlphaGo -- Part 1

AlphaGo is a computer program that plays the board game Go. It was developed by the London-based DeepMind Technologies, an acquired subsidiary of Google. Subsequent versions of AlphaGo became increasingly powerful, including a version that competed under the name Master. After retiring from competitive play, AlphaGo Master was succeeded by an even more powerful version known as AlphaGo Zero, which was completely self-taught without learning from human games. AlphaGo Zero was then generalized into a program known as AlphaZero, which played additional games, including chess and shogi. AlphaZero has in turn been succeeded by a program known as MuZero which learns without being taught the rules. AlphaGo and its successors use a Monte Carlo tree search algorithm to find its moves based on knowledge previously acquired by machine learning, specifically by an artificial neural network (a deep learning method) by extensive training, both from human and computer play. A neural network is trained to identify the best moves and the winning percentages of these moves. This neural network improves the strength of the tree search, resulting in stronger move selection in the next iteration. In October 2015, in a match against Fan Hui, the original AlphaGo became the first computer Go program to beat a human professional Go player without handicap on a full-sized 19×19 board. In March 2016, it beat Lee Sedol in a five-game match, the first time a computer Go program has beaten a 9-dan professional without handicap. Although it lost to Lee Sedol in the fourth game, Lee resigned in the final game, giving a final score of 4 games to 1 in favour of AlphaGo. In recognition of the victory, AlphaGo was awarded an honorary 9-dan by the Korea Baduk Association. The lead up and the challenge match with Lee Sedol were documented in a documentary film also titled AlphaGo, directed by Greg Kohs. The win by AlphaGo was chosen by Science as one of the Breakthrough of the Year runners-up on 22 December 2016. At the 2017 Future of Go Summit, the Master version of AlphaGo beat Ke Jie, the number one ranked player in the world at the time, in a three-game match, after which AlphaGo was awarded professional 9-dan by the Chinese Weiqi Association. After the match between AlphaGo and Ke Jie, DeepMind retired AlphaGo, while continuing AI research in other areas. The self-taught AlphaGo Zero achieved a 100–0 victory against the early competitive version of AlphaGo, and its successor AlphaZero was perceived as the world's top player in Go by the end of the 2010s.

AlphaGo -- Part 2

AI community AlphaGo's March 2016 victory was a major milestone in artificial intelligence research. Go had previously

been regarded as a hard problem in machine learning that was expected to be out of reach for the technology of the time. Most experts thought a Go program as powerful as AlphaGo was at least five years away; some experts thought that it would take at least another decade before computers would beat Go champions. Most observers at the beginning of the 2016 matches expected Lee to beat AlphaGo. With games such as checkers (that has been solved by the Chinook computer engine), chess, and now Go won by computers, victories at popular board games can no longer serve as major milestones for artificial intelligence in the way that they used to. Deep Blue's Murray Campbell called AlphaGo's victory "the end of an era... board games are more or less done and it's time to move on." When compared with Deep Blue or Watson, AlphaGo's underlying algorithms are potentially more general-purpose and may be evidence that the scientific community is making progress towards artificial general intelligence. Some commentators believe AlphaGo's victory makes for a good opportunity for society to start preparing for the possible future impact of machines with general purpose intelligence. As noted by entrepreneur Guy Suter, AlphaGo only knows how to play Go and doesn't possess general-purpose intelligence; "[It] couldn't just wake up one morning and decide it wants to learn how to use firearms." AI researcher Stuart Russell said that AI systems such as AlphaGo have progressed quicker and become more powerful than expected, and we must therefore develop methods to ensure they "remain under human control". Some scholars, such as Stephen Hawking, warned (in May 2015 before the matches) that some future self-improving AI could gain actual general intelligence, leading to an unexpected AI takeover; other scholars disagree: AI expert Jean-Gabriel Ganascia believes that "Things like 'common sense'... may never be reproducible", and says "I don't see why we would speak about fears. On the contrary, this raises hopes in many domains such as health and space exploration." Computer scientist Richard Sutton said "I don't think people should be scared... but I do think people should be paying attention." In China, AlphaGo was a "Sputnik moment" which helped convince the Chinese government to prioritize and dramatically increase funding for artificial intelligence. In 2017, the DeepMind AlphaGo team received the inaugural IJCAI Marvin Minsky medal for Outstanding Achievements in AI. "AlphaGo is a wonderful achievement, and a perfect example of what the Minsky Medal was initiated to recognise", said Professor Michael Wooldridge, Chair of the IJCAI Awards Committee. "What particularly impressed IJCAI was that AlphaGo achieves what it does through a brilliant combination of classic AI techniques as well as the state-of-the-art machine learning techniques that DeepMind is so closely associated with. It's a breathtaking demonstration of contemporary AI, and we are delighted to be

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able to recognise it with this award."

AlphaGo -- Part 3

Style of play Toby Manning, the match referee for AlphaGo vs. Fan Hui, has described the program's style as "conservative". AlphaGo's playing style strongly favours greater probability of winning by fewer points over lesser probability of winning by more points. Its strategy of maximising its probability of winning is distinct from what human players tend to do which is to maximise territorial gains, and explains some of its odd-looking moves. It makes a lot of opening moves that have never or seldom been made by humans. It likes to use shoulder hits, especially if the opponent is over concentrated.